

Model Question Paper I**MACHINE LEARNING***Time: 3 Hour**Max. Marks: 75***PART A**

- I. Answer **all** questions in one word or one sentence. Each question carries **1 mark**.
(9 * 1 = 9 marks)

1	List any two characteristics of Reinforcement Machine Learning	M1.01	R
2	List any two advantages of dimensionality reduction	M1.04	R
3	List the task of Machine learning	M1.02	R
4	List four different types of Clustering techniques	M2.04	R
5	List any two applications of Clustering Machine Learning model	M2.04	R
6	List Pruning methods	M3.02	R
7	What is Cross Entropy	M3.01	R
8	List applications of CNN	M4.04	R
9	List characteristics of a Perceptron	M4.01	R

PART B

1. Answer any **eight** questions from the following, each question carries **3 marks**.
(8 * 3=24 marks)

1	Differentiate Model Parameter and Hyperparameter	M1.04	U
2	Explain Feature scaling techniques	M1.03	U
3	Compare Supervised and Unsupervised Machine Learning	M1.01	U
4	Explain Gradient Descent with example	M2.02	U
5	Compare the cost functions - Mean Square Error and Residual	M2.01	U
6	Compare Regularized Linear regression models - Ridge and Lasso	M2.03	U
7	List advantages and disadvantages of Logistic Regression	M3.01	U
8	Explain Naïve Bayes Classification Model	M3.03	U
9	Explain any two Pooling Functions	M4.01	U
10	List the benefits of transfer learning	M4.04	R

PART C

Answer **ALL** questions. Each question carries **7 marks**.

(6*7=42 marks)

III	Explain Principal Component Analysis	M1.02	U
OR			
IV	Explain Data preprocessing	M1.03	U
V	Explain Performance measures using confusion matrix	M1.02	U
OR			
VI	Explain any four Machine Learning challenges	M1.04	U
VII	Explain Gradient Descent	M2.02	U
OR			
VIII	Explain K-Means Clustering model	M2.05	U
IX	Explain Support Vector Machine	M3.04	U
OR			
X	Explain Naïve Bayes Classification Model	M3.03	U
XI	Explain Logistic Regression	M3.01	U
OR			
XII	Explain Random Decision forest	M3.02	U
XIII	Explain Convolution operation with example	M4.02	A
OR			
XIV	Explain activation functions in ANN	M4.03	U

